Context-Sensitive Pragmatic Classification for Burmese: A 4-Stage Ablation Study with End-to-End Pipeline

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Abstract

We present a system for classifying the pragmatic politeness level of Burmese utterances using a fine-tuned multilingual encoder model. Burmese is a low-resource language in which pragmatic meaning is highly sensitive to social context, speaker hierarchy, and linguistic register. We design a four-stage ablation study to systematically investigate how progressively richer contextual signals affect classification quality. In Stage 1, the model receives only the raw utterance. Stage 2 adds social context and communicative instruction. Stage 3 provides explicit pragmatic metadata (register, power relation, tone) as a text prefix serving as an oracle upper bound. Stage 4 closes the loop by replacing the oracle metadata with automatic predictions from three independently trained auxiliary classifiers, forming a fully end-to-end pipeline. Our results show that contextual signals improve macro F1 by +0.057 (Stage 1→2) and oracle metadata adds a further +0.041 (Stage 2→3). However, the end-to-end pipeline incurs a −0.097 drop, demonstrating a significant error propagation cost when metadata is predicted rather than given. We additionally contribute a theoretically grounded tone taxonomy derived from Appraisal Theory, mapping 131 informal annotations to 5 interpretable categories. All models and an interactive demo are publicly available on HuggingFace.

1 Introduction

Natural language understanding requires more than recognising the literal meaning of words. Pragmatics the study of how context shapes meaning is central to social language use. In Burmese, a morphologically and pragmatically complex low-resource language, the same underlying intent can be realised through dozens of distinct utterance forms depending on the social relationship between speakers, the formality of the situation, and the speaker's emotional stance. Correctly identifying these pragmatic properties is a prerequisite for socially aware NLP systems.

Prior work on controllable text generation in Burmese has explored prompting multilingual sequence-to-sequence models with structured attribute inputs (Xue et al., 2021). However, this generation framing presents evaluation challenges: when many surface forms are all valid realisations of the same intent, metrics such as BLEU are unreliable. We argue that a classification framing predicting the pragmatic label of an utterance provides a more tractable and measurable research problem for the same underlying phenomenon.

This paper makes the following contributions:

- We propose a four-stage ablation design for pragmatic classification in Burmese, systematically varying the richness of contextual input to a fine-tuned `xlm-roberta-base` model.
- We introduce three auxiliary classifiers that predict register, power relation, and tone from utterance text, enabling a fully automatic end-to-end pipeline (Stage 4) requiring no manual metadata.
- We quantify the error propagation cost of the pipeline (−0.097 macro F1), identifying intermediate classifier accuracy as the key bottleneck.
- We propose a theoretically grounded tone taxonomy grounded in Appraisal Theory (Martin and White, 2005), mapping 131 informal annotations to 5 interpretable categories and providing an academically defensible labelling scheme for low-resource pragmatic data.
- We release all seven fine-tuned models and an interactive demo on HuggingFace.

2 Research Questions

This study is structured around three research questions:

RQ1: How much does adding social context and communicative intent to the model input improve Burmese politeness classification compared to an utterance-only baseline?

RQ2: What is the oracle upper bound achievable when explicit pragmatic metadata (register, power relation, tone) is provided as input?

RQ3: How much of the oracle benefit is preserved when pragmatic metadata is predicted automatically, and what is the error propagation cost of the end-to-end pipeline?

3 Related Work

3.1 Multilingual Pre-trained Models

XLM-RoBERTa (Conneau et al., 2020) is a cross-lingual encoder trained on 2.5TB of filtered CommonCrawl text across 100 languages, including Burmese. Its encoder-only architecture produces a single contextualised representation that is well-suited for classification tasks. We adopt XLM-RoBERTa-base as our backbone, as it is purpose-built for natural language understanding rather than generation.

The mT5 model (Xue et al., 2021) extends the T5 framework to a massively multilingual setting and has been used for generation-based approaches to Burmese pragmatics. We use mT5 as a baseline point of comparison in framing the task.

3.2 Controlled and Pragmatic NLP

Controllable text generation has explored various mechanisms for conditioning outputs on stylistic or pragmatic attributes (See et al., 2019; Madotto et al., 2020). However, most such work focuses on high-resource languages and coarse attributes such as sentiment or topic. Context-aware dialogue generation has similarly examined politeness and social factors (Li et al., 2016; Rashkin et al., 2019), but rarely in low-resource settings.

For low-resource languages specifically, the interaction between limited data, label quality, and pragmatic complexity presents unique challenges. Our work addresses this by proposing a label coarsening strategy grounded in established sociolinguistic theory.

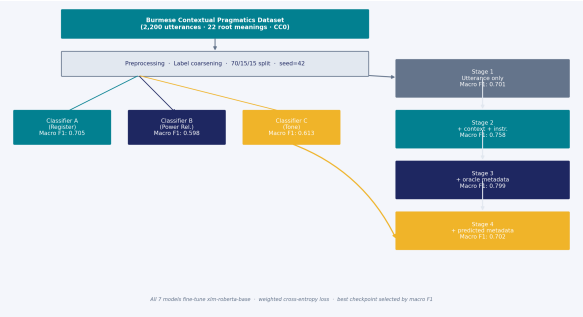

Figure 1: System overview. The dataset feeds into preprocessing and then into seven independently fine-tuned models: three auxiliary classifiers (A, B, C) predicting register, power relation, and tone respectively, and four politeness classifiers covering Stages 1–4.

3.3 Appraisal Theory

Appraisal Theory (Martin and White, 2005) provides a systematic framework for analysing evaluative language. Its Attitude subsystem distinguishes Affect (emotional states), Judgement (social evaluations), and Appreciation (value assessments). We use this framework to map 131 informal tone annotations from our dataset into 5 principled categories, addressing a key annotation quality limitation.

3.4 Error Propagation in NLP Pipelines

Pipeline architectures, where one model's output feeds into the next, are well-known to suffer from error propagation (Finkel et al., 2006). Our Stage 3→Stage 4 analysis contributes a concrete quantification of this effect in the context of pragmatic classification for a low-resource language.

4 Dataset

We use the Burmese Contextual Pragmatics dataset (freococo, 2024), a publicly available CC0-licensed resource containing 2,200 Burmese utterances covering 22 root conversational meanings (e.g., "I love you", "Thank you", "Stop it"). Each root meaning has approximately 100 variations realised across different social registers, politeness levels, and situational contexts. An overview of the full system is shown in Figure 1.

Each example provides: the Burmese utterance, a social context description, a communicative instruction, register (colloquial / standard / formal / slang), politeness (13 original labels), power relation (8 original values), tone (131 free-text values), and additional fields including emotion, intent strength, and phonetic transcription.

Original label	Mapped to
neutral, polite, informal, professional, blunt, rude	(keep as-is)
very_polite, religious, polite_but_firm	polite
friendly	informal
impolite, sarcastic	rude
stern	blunt

Table 1: Politeness label coarsening: 13 original classes collapsed to 6.

Category	Representative original values
neutral	Neutral, Direct, Objective, Indifferent
positive	Warm, Sweet, Cheerful, Sincere, Hopeful
formal	Professional, Serious, Respectful, Humble
negative	Angry, Hostile, Harsh, Disgusted, Sarcastic
emotional	Urgent, Pleading, Sad, Panicked, Apologetic

Table 2: Tone taxonomy grounded in Appraisal Theory (Martin and White, 2005).

The dataset was constructed by a single native speaker without a formal annotation protocol, which introduces quality issues particularly in the tone field, which contains typos (e.g., *Appologetic*), compound labels (e.g., *Nervous/Sincere*), and singleton values (e.g., *Spooky*, *Whispering*). We address these issues through principled preprocessing described in Section 5.

5 Preprocessing

5.1 Politeness Label Coarsening

The original politeness field contains 13 labels, several of which have very few instances (e.g., *sarcastic*: 1, *stern*: 2). We merge semantically similar labels into 6 learnable classes using the mapping in Table 1.

5.2 Tone Taxonomy via Appraisal Theory

The tone field presents a more significant challenge: 131 distinct free-text values assigned informally by a single annotator. These are unlearnable as classification targets and lack academic justification.

We address this by mapping the 131 values to 5 categories grounded in the Attitude subsystem of Appraisal Theory (Martin and White, 2005), as shown in Table 2. This is a standard practice in low-resource NLP research where annotation quality is variable (Plank, 2022).

5.3 Power Relation Coarsening

Eight original power relation values are collapsed to four by merging role-specific variants of the inferior relation into a single *inferior_to_superior* class: *customer_to_staff*, *customer_to_seller*, *passenger_to_driver*, and *patient_to_doctor* all reflect the same fundamental speaker hierarchy dynamic.

5.4 Class Imbalance

The politeness label is severely imbalanced: *neutral* constitutes 53% of all examples. Without correction, a model trained with standard cross-entropy loss would learn to predict *neutral* for all inputs, achieving 53% accuracy while failing on all other classes.

We address this with weighted cross-entropy loss, assigning each class a weight of $\frac{N}{K \cdot n_k}$ where N is the total training examples, K is the number of classes, and n_k is the count of class k. Class weights range from 0.318 (neutral) to 6.111 (rude). We report macro F1 as our primary evaluation metric, which treats all classes equally regardless of support.

5.5 Data Splits

We split the dataset 70/15/15 (train/validation/test) using a fixed random seed of 42. The same split is used consistently across all seven models trained in this work. All reported results are from the held-out test set (330 examples).

6 Methodology

6.1 Base Model

We fine-tune `xlm-roberta-base` (Conneau et al., 2020) for all classification tasks. XLM-RoBERTa is an encoder-only transformer trained on 100 languages including Burmese, making it appropriate for understanding Burmese pragmatic text. Its classification head consists of a linear layer over the `[CLS]` token representation.

6.2 Training Setup

All models are trained with the following hyperparameters: 8 epochs, batch size 8, learning rate 2×10^{-5}, warmup steps 150, weight decay 0.01, weighted cross-entropy loss. We use HuggingFace `Trainer` with a custom `WeightedTrainer` subclass that overrides `compute_loss` to inject class-weighted loss. The best checkpoint is selected by macro F1 on the validation set using

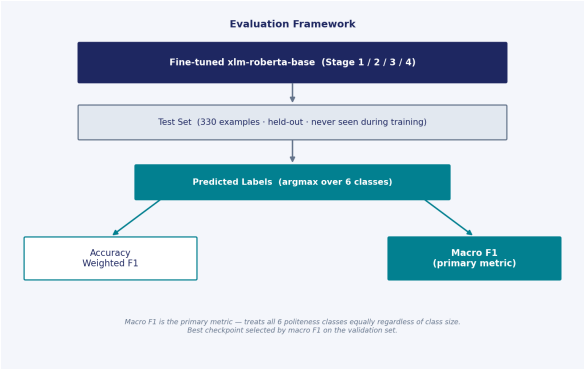

Figure 2: Evaluation framework. All models are evaluated on the held-out test set (330 examples) using accuracy, weighted F1, and macro F1. Macro F1 is the primary metric as it treats all six politeness classes equally regardless of class size.

`load_best_model_at_end=True`. The evaluation framework is shown in Figure 2.

6.3 Four-Stage Ablation Design

Our main experimental contribution is a four-stage ablation over the input format given to the politeness classifier. The architecture, training procedure, and evaluation setup are identical across all stages; only the input text changes. This design directly answers RQ1–3.

Stage 1 Utterance only. The model receives only the raw Burmese utterance. This is the baseline, answering whether XLM-RoBERTa can classify Burmese pragmatic properties at all.

Stage 2 Utterance + context + instruction. The social context and communicative instruction are appended as separate segments using the XLM-RoBERTa separator token:

```
[utterance] </s> [context] </s>
[instruction]
```

This tests whether knowing the social situation improves classification.

Stage 3 Oracle metadata prefix. Ground-truth register, power relation, and tone labels are prepended as readable text tags:

```
[register: X] [power: Y] [tone:
Z] [utterance] </s> [context] </s>
[instruction]
```

This is the theoretical upper bound assuming perfect metadata.

Stage 4 Predicted metadata prefix (end-to-end pipeline). The same input format as Stage 3, but metadata values are replaced with predictions from Classifiers A, B, and C respectively. No human-provided metadata is required at inference time.

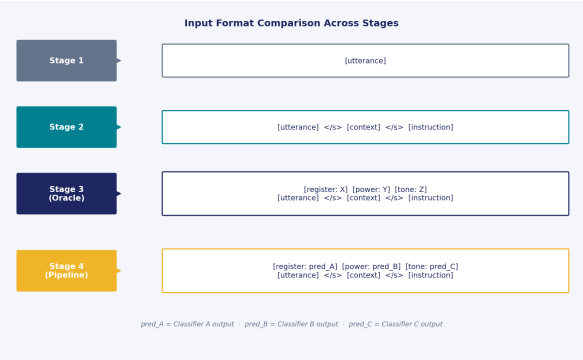

Figure 3: Input format comparison across the four stages. Each stage adds progressively richer contextual information to the model input.

Clf.	Task	Acc.	Macro F1	Hardest class
A	Register	0.739	0.705	standard (0.53)
B	Power relation	0.739	0.598	any (0.40)
C	Tone	0.667	0.613	negative (0.41)

Table 3: Intermediate classifier results on the test set.

Figure 3 illustrates the input format used at each stage.

6.4 Auxiliary Classifiers for Stage 4

To enable Stage 4, we train three independently fine-tuned classifiers on the same dataset and split.

Classifier A (Register) takes only the utterance as input and predicts register $\in$ {colloquial, standard, formal, slang}. Register is largely encoded in the text itself through vocabulary and sentence-final particles.

Classifier B (Power relation) takes utterance plus context and predicts power relation $\in$ {equal, inferior_to_superior, superior_to_inferior, any}. Context is required because power relation is a social rather than purely linguistic property.

Classifier C (Tone) takes utterance plus context and predicts coarsened tone $\in$ {neutral, positive, formal, negative, emotional}.

Figure 4 shows the full end-to-end flow of the Stage 4 pipeline.

7 Results

7.1 Intermediate Classifiers

Table 3 reports test-set results for the three auxiliary classifiers.

Classifier A achieves the strongest performance. *colloquial* (F1=0.82) and *formal* (F1=0.79) are

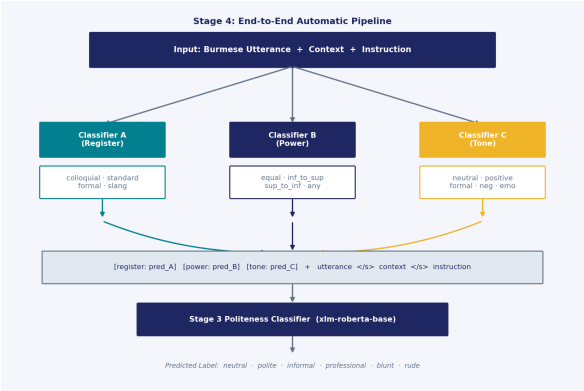

Figure 4: Stage 4 end-to-end pipeline. Three auxiliary classifiers predict register, power relation, and tone automatically; their outputs are assembled into a text prefix that is fed into the Stage 3 politeness classifier.

Stage	Input	Acc.	Macro F1	Δ Macro F1
Stage 1	Utterance only	0.739	0.701	—
Stage 2	+ context + instruction	0.794	0.758	+0.057
Stage 3	+ oracle metadata	0.846	0.799	+0.041
Stage 4	+ predicted metadata	0.727	0.702	−0.097

Table 4: Full ablation results. Macro F1 is the primary metric.

well-separated, but *standard* (F1=0.53) remains difficult due to its ambiguous middle-ground position between registers. Classifier B struggles with *any* (F1=0.40) and *superior_to_inferior* (F1=0.43), both of which are underrepresented and semantically difficult. Classifier C achieves moderate performance; *negative* (F1=0.41) has only 17 test examples.

7.2 Politeness Classification: Main Ablation

Table 4 presents the main ablation results.

7.3 Per-Class Results: Stage 3

Table 5 shows per-class results for Stage 3, the best-performing model.

professional achieves the highest F1 (0.94) with perfect recall, reflecting the distinctiveness of formal Burmese vocabulary. *blunt* shows an interesting asymmetry: precision of 1.00 but recall of 0.50, indicating the model is conservative when it predicts *blunt* it is always correct, but it misses half the blunt examples. *rude* and *blunt* are the weakest classes, both with 12–14 test examples.

7.4 Per-Class Results: Stage 4

Table 6 shows per-class results for Stage 4 for comparison with Stage 3.

Class	Prec.	Rec.	F1	Support
blunt	1.00	0.50	0.67	14
informal	0.81	0.85	0.83	59
neutral	0.89	0.86	0.87	174
polite	0.75	0.91	0.82	56
professional	0.88	1.00	0.94	15
rude	0.78	0.58	0.67	12
macro avg	0.85	0.78	0.80	330

Table 5: Per-class results for Stage 3 (oracle metadata).

Class	Prec.	Rec.	F1	Support
blunt	0.89	0.57	0.70	14
informal	0.72	0.69	0.71	59
neutral	0.88	0.68	0.77	174
polite	0.53	0.91	0.67	56
professional	0.83	1.00	0.91	15
rude	0.43	0.50	0.46	12
macro avg	0.71	0.73	0.70	330

Table 6: Per-class results for Stage 4 (predicted metadata pipeline).

8 Analysis

RQ1: Does context help? Yes. Stage 2 improves over Stage 1 by +0.057 macro F1, confirming that social context and communicative instruction carry real pragmatic signal. The utterance alone is insufficient for reliable politeness classification, which is linguistically expected: Burmese politeness marking depends heavily on speaker-listener relationship and situational formality.

RQ2: What is the oracle upper bound? Stage 3 achieves 0.799 macro F1, a further +0.041 over Stage 2. When the model is given exact register, power relation, and tone labels, its classification improves. This demonstrates that these pragmatic dimensions are genuinely informative for politeness prediction.

RQ3: How much does error propagation cost? Stage 4 drops to 0.702 macro F1, −0.097 below the oracle (Stage 3). Strikingly, Stage 4 achieves nearly identical performance to Stage 1 (0.701), despite having access to predicted metadata. This demonstrates that *noisy predicted metadata is almost as uninformative as no metadata at all*. The bottleneck is intermediate classifier accuracy: Classifiers A, B, and C achieve macro F1 of 0.705, 0.598, and 0.613 respectively. The errors in these classifiers, when propagated through the pipeline, negate the benefit of the metadata signal identified in Stage 3.

This finding is not a failure of the approach but a precise quantification of the error propagation problem in pragmatic pipeline systems. It directly motivates future work on improving the intermediate classifiers.

Value of the tone taxonomy. The Appraisal Theory mapping is an enabling contribution: without reducing 131 annotations to 5 principled categories, Classifier C could not be trained, and Stage 4 would be impossible. The mapping also makes the tone labels interpretable and academically citable, addressing a key weakness of the original dataset.

9 Direction Change: From Generation to Classification

This project originally proposed a generation framing: given structured pragmatic attributes, generate the appropriate Burmese utterance using mT5. We pivoted to classification for three reasons. First, the dataset contains only 22 root meanings with approximately 100 near-identical surface variations each, providing insufficient diversity for a generation model to generalise. Second, BLEU and chrF are unreliable when multiple surface forms are valid realisations of the same intent. Third, early training runs confirmed non-convergence: training loss remained at 27+ over 8 epochs, and one run produced `training_loss=0.0, validation_loss=NaN` due to a GPU FP16 precision issue.

Classification with XLM-RoBERTa provides falsifiable research claims, clean metrics, and a well-defined evaluation setup. The research questions shift from "can the model generate appropriate utterances?" to "can the model identify pragmatic properties of utterances?", which is an equally valid and more measurable NLP problem for this dataset.

10 Summary of Progress

All training, evaluation, and deployment work is complete. Table 7 summarises the status of each component.

Component	Status
Dataset preprocessing + label coarsening	Done
Tone taxonomy (Appraisal Theory)	Done
Train/val/test split (seed=42)	Done
Classifier A: register (macro F1: 0.705)	Done
Classifier B: power relation (0.598)	Done
Classifier C: tone (0.613)	Done
Stage 1: utterance only (0.701)	Done
Stage 2: + context (0.758)	Done
Stage 3: oracle metadata (0.799)	Done
Stage 4: end-to-end pipeline (0.702)	Done
All 7 models on HuggingFace Hub	Done
Interactive demo Spaces (v1 + v2)	Done
Zero-shot baseline (GPT-4, Gemini)	Next week
Intermediate classifier improvement	Next week
Error analysis with Burmese examples	Next week
Final paper: methodology + results	Next week

Table 7: Project progress summary.

11 Limitations

This study has several limitations. The dataset is small (2,200 examples) and covers only 22 root meanings, which constrains generalisability. The tone taxonomy, while theoretically grounded, was applied manually and a future annotation study could verify inter-annotator agreement on the coarsened categories. The evaluation of the end-to-end pipeline is limited to a single test split; cross-validation would provide more stable estimates of the error propagation cost. Finally, all experiments use a single base model (`xlm-roberta-base`); larger models or Burmese-specific pre-trained models may yield different trade-offs.

12 Conclusion

We have presented a four-stage ablation study for Burmese pragmatic politeness classification, extending from a simple utterance-only baseline to a fully end-to-end pipeline with automatic metadata prediction. Our results show that social context is a meaningful signal (+0.057 macro F1), oracle pragmatic metadata provides a further gain (+0.041), and the end-to-end pipeline incurs a significant error propagation cost (-0.097) that nearly eliminates the metadata benefit. We contribute a theoretically grounded tone taxonomy and release all models publicly. Future work will focus on improving the intermediate classifiers to close the oracle-pipeline gap, conducting a zero-shot comparison against large language models such as GPT-4 and Gemini, and performing error analysis on misclassified Burmese examples.

References

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettle-

moyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.

Jenny Rose Finkel, Christopher D. Manning, and Andrew Y. Ng. 2006. Solving the problem of cascading errors: Approximate Bayesian inference for linguistic annotation pipelines. In *Proceedings of the 2006 Conference on Empirical Methods in Natural Language Processing*, pages 618–626, Sydney, Australia. Association for Computational Linguistics.

freococo. 2024. Burmese contextual pragmatics dataset. `https://huggingface.co/datasets/freococo/burmese-contextual-pragmatics`. Accessed: 2026.

Jiwei Li, Michel Galley, Chris Brockett, Georgios Spithourakis, Jianfeng Gao, and Bill Dolan. 2016. A persona-based neural conversation model. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 994–1003, Berlin, Germany. Association for Computational Linguistics.

Andrea Madotto, Etsuko Ishii, Zhaojiang Lin, Sumanth Dathathri, and Pascale Fung. 2020. Plug-and-play conversational models. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 2422–2433, Online. Association for Computational Linguistics.

J.R. Martin and P.R.R. White. 2005. *The Language of Evaluation: Appraisal in English*. Palgrave Macmillan, London.

Barbara Plank. 2022. The "problem" of human label variation: On ground truth in data, modeling and evaluation. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 10671–10682, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Hannah Rashkin, Eric Michael Smith, Margaret Li, and Y-Lan Boureau. 2019. Towards empathetic open-domain conversation models: A new benchmark and dataset. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5370–5381, Florence, Italy. Association for Computational Linguistics.

Abigail See, Stephen Roller, Douwe Kiela, and Jason Weston. 2019. What makes a good conversation? how controllable attributes affect human judgments. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 1702–1723, Minneapolis, Minnesota. Association for Computational Linguistics.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. mT5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 483–498, Online. Association for Computational Linguistics.